

Zach Taylor

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EXPERIENCE

Software Engineer Intern *Rohde & Schwarz — C++, Python*

April 2024 – Jan 2025

- Created tailored network analysis solutions using **C++** for client specific needs along with enabling faster diagnostics and informed decision making.
- Refactored slow, inefficient code to boost long-term maintainability as well as integration into new and existing software products.
- Discovered **15** critical computation errors affecting all software applications through **unit testing**, resolving the issue by designing more effective **APIs**.
- Automated repetitive tasks with **Python**, shortening development time and improving team productivity.

Software Development Intern *Portland State University — Python, TensorFlow*

June 2019 – September 2019

- To increase efficiency in food processing and make use of AI image recognition, created a **TensorFlow** neural network model using **Python** with **OpenCV**.
- Optimized the model using a transfer learning approach, resulting in a **97%** detection accuracy through measured tests on factory assembly lines.
- Coordinated weekly meetings with engineers as well as end users to align product direction with objectives and keep development on schedule.

Undergraduate Teaching Assistant *Oregon State University*

September 2020 – June 2023

- Delivered personalized one-on-one and small group tutoring focused on reinforcing core concepts and addressing individual challenges, raising academic performance by **30%** among struggling students.
- Taught fundamental software courses including data structure and operating systems using **C** as well as full stack web development with **JavaScript**.
- Proctored exams containing **100-150** students, upholding academic integrity and ensuring a fair, distraction free testing environment.

PROJECTS

Patient Monitoring Robot | *Python, OpenCV, AI*

[GitHub](#)

- Developed an **AI** tracking robot to reduce falls among elderly patients in hospitals and nursing homes, utilizing **Python**, **OpenCV**, and **MediaPipe** libraries.
- Increased the fall detection accuracy by **56%** through live model training in local nursing homes along with creating custom MediaPip functions for figure tracking.
- Directed development efforts in a team of **7-8** engineers by mentoring members on AI deployment and performance optimization.
- Displayed progress to Intel sponsors during weekly meetings and presented the project at the 2023 OSU engineering expo.

SIMD Image Filter | *C++, SIMD*

[GitHub](#)

- Implemented a **C++** image filter library to learn the fundamentals of **SIMD** uses cases and architecture.
- Features blur, sharpen, and edge detection algorithms with metrics to show a **5x-7x** difference between SIMD and scalar applications.

EDUCATION

Oregon State University - Bachelor of Science

September 2019 – June 2023

Winner of the Pioneering Concept Award at the 2023 OSU Engineering Expo

SKILLS

C++, Python, Typescript, Next.js, TensorFlow, CUDA, GitHub, Linux, Jenkins, Visual Studio